

Yene Irvine

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TECHNICAL SKILLS

Data Engineering: SQL, Data warehousing (Snowflake, DB2), ETL / ELT pipelines
Backend: Node.js (Express, NestJS), FastAPI, Java, PostgreSQL, SQLAlchemy, REST & SOAP APIs
Frontend: TypeScript, JavaScript (React), Tailwind CSS, Redux, Webpack, Babel
Cloud & DevOps: AWS, Azure, GCP, Vercel, Docker, Terraform, Redis, GitHub Actions, RabbitMQ, CI/CD pipelines
Machine Learning: PyTorch, scikit-learn, Pandas, NumPy, matplotlib, Weights and Biases (WandB), HPC compute

EXPERIENCE

- Data Engineer** Sep. 2024 – Present
Fuel Transport Inc. Montreal, QC
- Built and integrated a Snowflake Rate Editor (FastAPI, React) to enable real-time editing of client rates, powering 100s of daily API calls and contributing to \$1.5M+ in revenue over 5 months.
 - Developed and implemented a custom Azure Data Factory pipeline to migrate data from IBM DB2 to Snowflake, replacing Fivetran and saving \$40K+ annually in third-party costs.
 - Optimized CI/CD pipelines (GitHub Actions, Docker, Azure Container Apps) to streamline deployment of the Snowflake Rate Editor's frontend and backend containers, improving reliability and reducing overhead.
 - Leveraged Azure Monitor, Application Insights, and related Azure services to track API performance, analyze traffic patterns, and proactively resolve issues, ensuring high availability for enterprise clients.
- Controls Engineer** Apr. 2019 – Jan. 2022
MCA Consultants Charlottetown, PE & Halifax, NS
- Developed and commissioned 25 control narratives for utilities and HVAC control systems, integrating PLCs and process equipment, and working with contractors to enhance the Building Automation System.
 - Created a program in Excel to determine relieving requirements for pressurized vessels based on NFPA and ASME standards; used in designing venting configurations for 50+ major vessels.
 - Led a team to commission all process, utility, and HVAC systems and associated equipment under budget.

SELECTED PROJECTS

- RunIt** | *React Native, NestJS, PostgreSQL, TypeORM, Firebase Auth, JWT, Docker*
- Developed a cross-platform mobile app with Google sign-in (Firebase), real-time GPS run tracking, social connections, and challenge/profile features. Implemented a React Native frontend and NestJS backend with PostgreSQL (TypeORM), JWT session flow backed by Firebase Auth, and Dockerized deployment.
- AI DJ Setlister** | *Flask, Python, Spotify Web API, OAuth2, Genetic Algorithms, AWS EC2*
- Built a cloud-hosted Flask app allowing users to log in via Spotify OAuth2, select or paste playlists, and generate optimized DJ setlists using genetic algorithms that consider key compatibility, energy progression, and beat matching. Deployed on AWS EC2.
- Detecting Breathing Irregularities with Deep Learning** | *3D CNN, U-Net, PyTorch, HPC, WandB*
- Developed and deployed a 3D Convolutional Neural Network with University of Calgary Biometrics Lab to detect sleep-related medical emergencies via thermal video, leveraging HPC compute and WandB for real-time tracking at the Calgary Drop-In Centre.

EDUCATION

- University of Calgary** Calgary, AB
M.Eng., Software Engineering (GPA 3.94) Graduated May 2024
- Dalhousie University** Halifax, NS
B.Eng., Chemical Engineering (GPA 3.80) Graduated Apr. 2018

CERTIFICATIONS

AWS Certified Cloud Practitioner CLF-C02 (2024)
Microsoft Azure Fundamentals AZ-900 (2024)